Instrument Approach Procedure Chart Georeferencing

ACM Charting Group 26-01-411

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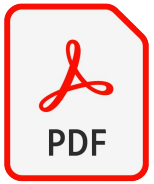
Mark Mentovai
mark@mentovai.com

Geospatial PDF

- First defined in a **2008 extension to PDF 1.7** (extension level 3, §8.8.1).
- Further standardized by **PDF 2.0** (ISO 32000-2:2020) §12.10, although PDF 2.0 is not widely adopted.
- Geospatial extension maps commonly used georeferencing concepts onto the PDF document structure.
  - **PROJCS**: projected coordinate system (such as Lambert Conformal Conic) and its parameters.
    - Embeds a **GEOGCS**: geographic coordinate system (such as NAD83/GRS80).
  - **GPTS**: georeferenced geographic (latitude–longitude) coordinates.
  - **LPTS**: corresponding page-centric Cartesian (x, y) coordinates.
  - **Bounds**: the neatline, or area within a page in which georeferencing is valid.
- Comparison to GeoTIFF.
  - First defined in 1995 (GeoTIFF 1.0).
  - Geospatial PDF PROJCS/GEOGCS correspond to GeoTIFF **GeoKey Directory** (GeoKeyDirectoryTag).
  - Geospatial PDF GPTS/LPTS correspond to GeoTIFF **tiepoints** (ModelTiepointTag)**.**
- **FAA-produced instrument approach charts in d-TPP have been georeferenced as Geospatial PDFs** starting in 2016, completed by 2017.
  - *Exception:* charts produced by **DoD/NGA** and distributed in TPP by FAA. These are bitmap (raster) images embedded in their PDFs, and are not georeferenced. These are all at military airfields or are high-altitude (military-only) procedures at civil/joint-use fields.
  - *Exception:* **charted visual flight procedures** (CVFP), like “Parkway Visual” and “River Visual”, are not georeferenced.
  - *Exception:* A small number of **continuation pages**.
  - *Exception:* KATL, KDTW, and KORD “**PRM AAUP**” notices, classified as IAPs because there’s no way to identify IAP AAUPs.
  - In cycle 2608, 10,010 out of 11,170 IAP PDFs (**90%**) are Geospatial PDFs.

Demonstration:
Open Geospatial PDF in Acrobat



→ open in →

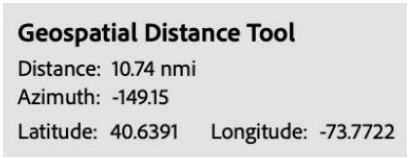
00610IL22L.PDF Adobe Acrobat

Steps

- Open a Geospatial PDF in Adobe Acrobat (Reader or Pro).
- All tools→(View more)→Measure objects.
- Select “Measuring tool” or “Geospatial location tool”.

Drawbacks

- No way to export data.
- **Not a real technique for programmatic data access.**
- This is just a demonstration showing that georeferencing data is in the PDF.



GDAL (Geospatial Data Abstraction Library)

- Open-source library and tools that operate on geospatial data formats...
  - ...including Geospatial PDF!
  - *and* GeoTIFF and many others, too.
- gdal.org
- Maintained by the Open Source Geospatial Foundation (OSGeo Project, osgeo.org).
- Command-line tools like `gdal info`.
- Library interfaces in many languages.
- Available for all major operating systems.
- MIT-licensed source code available at https://github.com/OSGeo/gdal.

Technique: Convert Geospatial PDF to GeoTIFF

Benefits

- GeoTIFF is a **mature container for georeferenced images**.
- Probably most the **broadly supported** format for this purpose.
- Existing tools and pipelines are **likely to interoperate** well with GeoTIFF.

Drawbacks

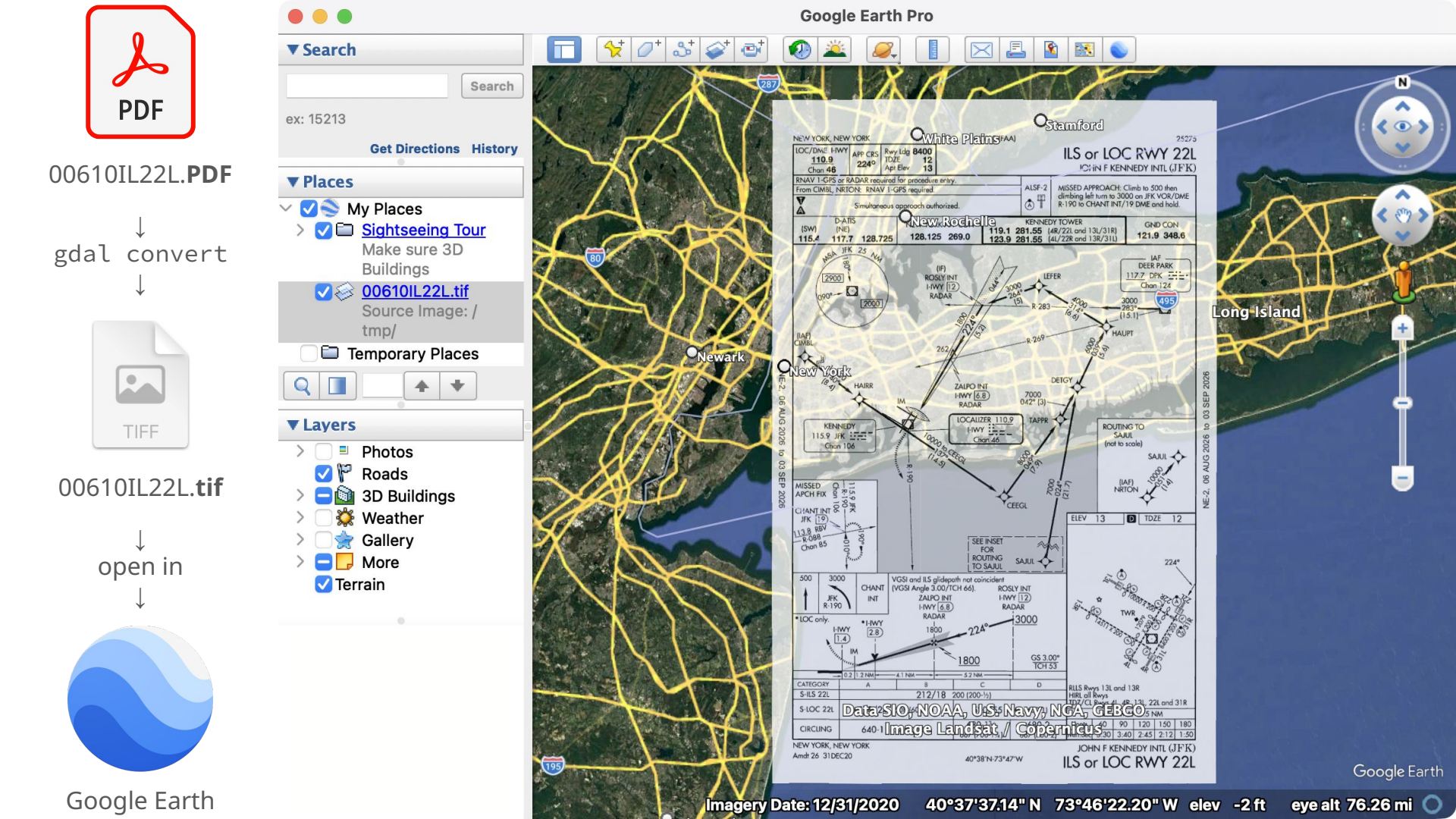
- Rasterization **loses valuable vector data** (text, clean rescaling, document structure) present in PDF files.
- **File sizes** generally exceed those of PDF, particularly at higher resolutions.
- Guidance:
  - *Downstream, data suppliers:* If you were going to rasterize before distribution anyway, these concerns won't apply.
  - *Upstream, FAA:*
    - I don't recommend pursuing this in place of the existing PDFs, because it constrains downstream uses.
    - However, there's some precedent for publishing a single product in multiple formats. d-VC (visual charts: sectionals, etc.) and IFR charts (enroute, etc.) are available in *both* georeferenced GeoTIFF and (non-georeferenced) PDF.

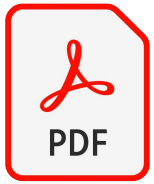
00610IL22L.**PDF** 00610IL22L.**tiff**

Technique: Convert Geospatial PDF to GeoTIFF

00610IL22L.**PDF** 00610IL22L.**tiff**

```
% gdal convert \
      --config GDAL_PDF_DPI=300 \
      --creation-option COMPRESS=DEFLATE \
      --creation-option ZLEVEL=9 \
      00610IL22L.PDF \
      00610IL22L.tiff
0...10...20...30...40...50...60...70...80...90...100 - done.
```

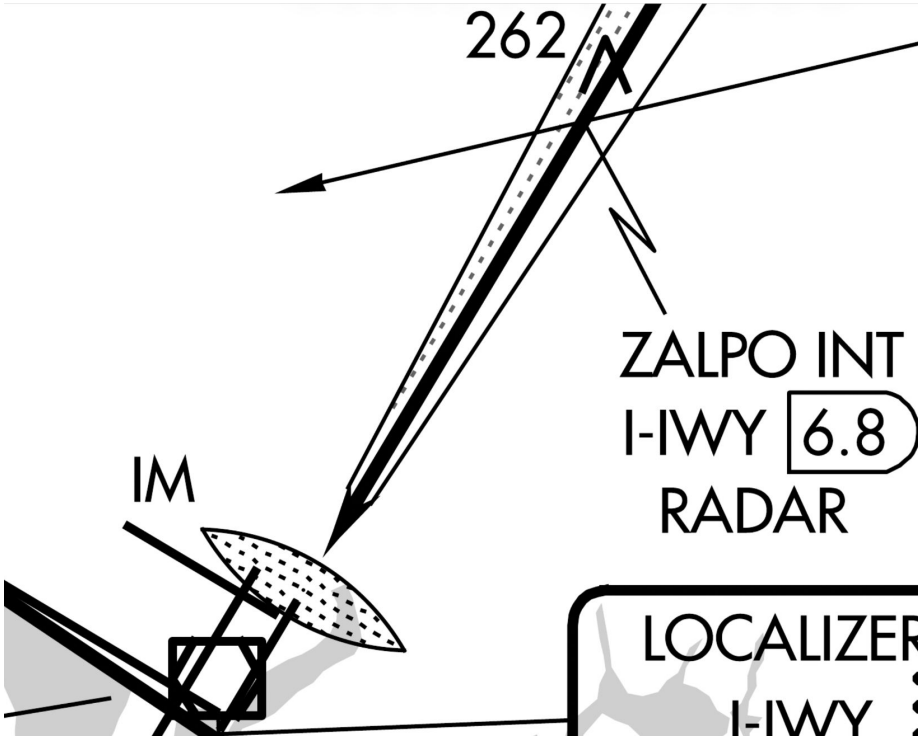

PDF
00610IL22L.PDF
gdal convert
TIFF
00610IL22L.tif
open in
Google Earth
Google Earth Pro
Search
Search
ex: 15213
Get Directions
History
Places
My Places
Sightseeing Tour
Make sure 3D Buildings
00610IL22L.tif
Source Image: / tmp/
Temporary Places
Layers
Photos
Roads
3D Buildings
Weather
Gallery
More
Terrain
Stamford
White Plains
New Rochelle
Newark
New York
Long Island
ILS or LOC RWY 22L
Data SIO, NOAA, U.S. Navy, NGA, GEBCO
Image Landsat / Copernicus
Google Earth
Imagery Date: 12/31/2020
40°37'37.14" N
73°46'22.20" W
elev -2 ft
eye alt 76.26 mi

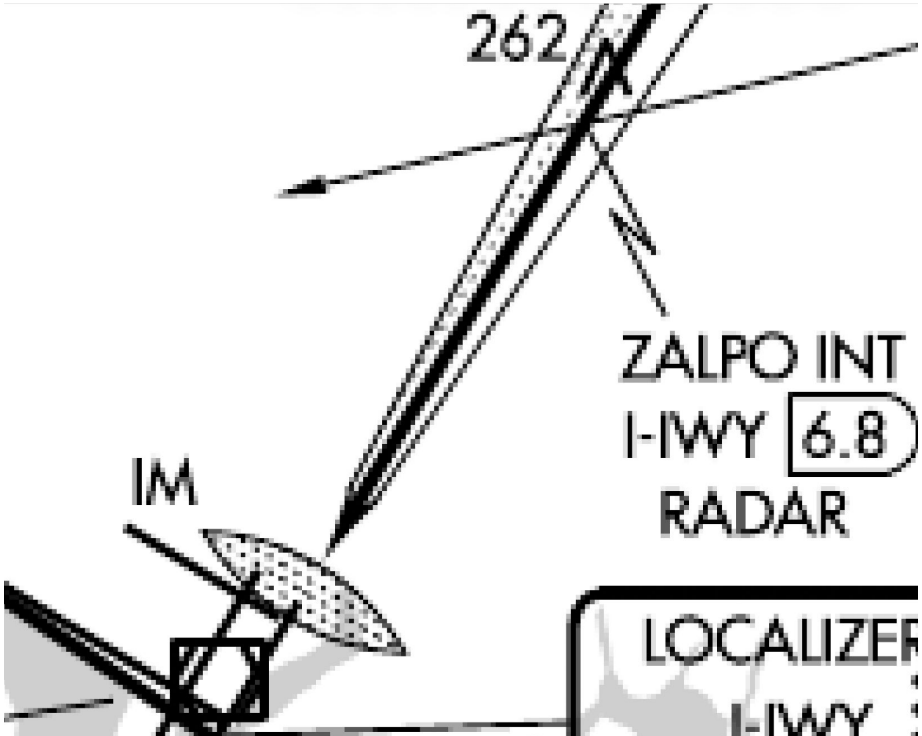


→ gdal convert →

00610IL22L.**PDF**
310kB
sharp when zoomed in

00610IL22L.**tiff**
2,930kB (uncompressed) @ 150dpi
pixelated when zoomed in





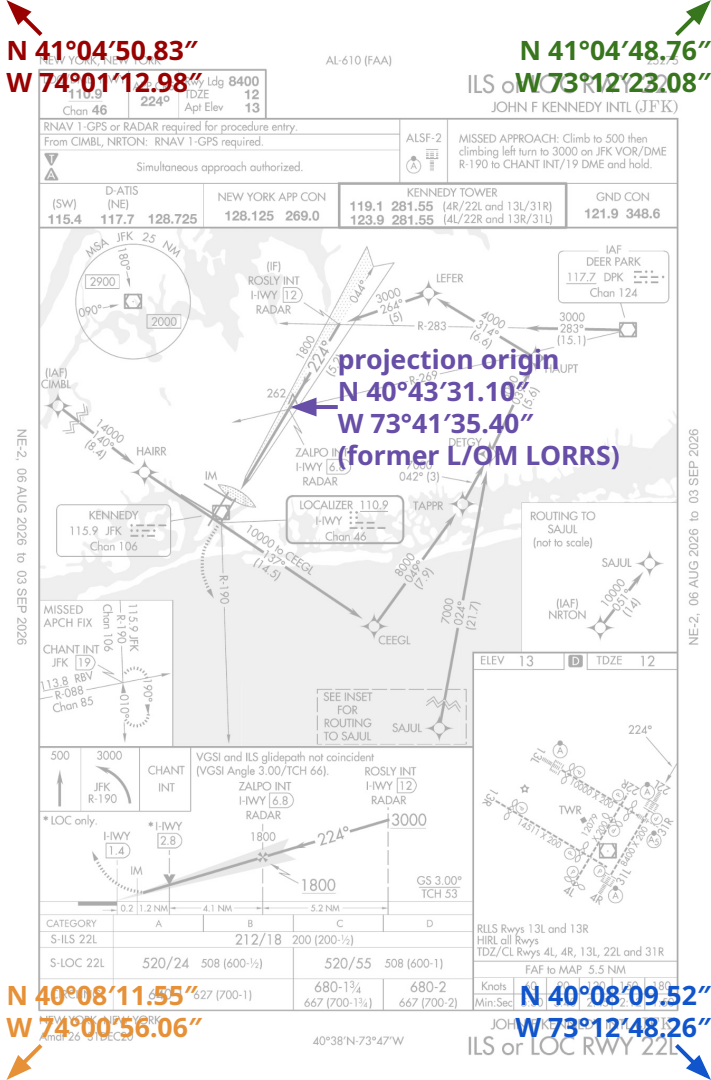
Technique: Extract Georeferencing Data from Geospatial PDF

- ...and put it *somewhere* in *some format.*
  - Figure that out later.
- This can be done upstream (FAA) and published, or it can be done downstream (data suppliers) for in-house use, using an identical technique.
- Parameters to extract and convey:
  - **Geographic coordinate system**, including datum and ellipsoid. *Always NAD83/GRS80 in FAA IAPs.*
  - Projected coordinate system. *Always Lambert Conformal Conic in FAA IAPs.* Requires parameters:
    - Two **standard parallels of latitude**.
    - **Origin coordinates**: latitude, and central meridian of longitude.
    - Offsets: false easting and northing. *Almost always 0 in practice; always 0 in FAA IAPs.*
  - **Control points**/tiepoints, relating geographic coordinates (latitude–longitude pairs) to page-centric Cartesian (x, y) positions on the chart, such as page corners.
- One-off use: run `gdal info`.
- Production pipeline:
  - **Script and scrape** `gdal info` (or `gdal info --format=json`).
  - **Use GDAL as a library**.
  - Other options are possible, addressed later.

Extract Georeferencing Data with `gdal info`

```
% gdal info --config GDAL_PDF_DPI=300 00610IL22L.PDF
Driver: PDF/Geospatial PDF
Files: 00610IL22L.PDF
Size is 1614, 2475
Coordinate Reference System WKT:
PROJCRS["",
    BASEGEOGCRS["NAD83",
        DATUM["North American Datum 1983",
            ELLIPSOID["GRS 1980",6378137,298.2572221,…],…],
        PRIMEM["Greenwich",0,…]],
    CONVERSION["unnamed",
        METHOD["Lambert Conic Conformal (2SP)",…],
        PARAMETER["Latitude of false origin",40.7253055555556,…],
        PARAMETER["Longitude of false origin",-73.6931666666667,…],
        PARAMETER["Latitude of 1st standard parallel",44.6666666666667,…],
        PARAMETER["Latitude of 2nd standard parallel",39.3333333333333,…],
        PARAMETER["Easting at false origin",0,…],
        PARAMETER["Northing at false origin",0,…]],
    CS[Cartesian,2],
        AXIS["(E)",east,ORDER[1],…],
        AXIS["(N)",north,ORDER[2],…]]
[…]
Corner Coordinates:
Upper Left  (-1081155.295, 1554888.755) ( 74d 1'12.98"W, 41d 4'50.83"N)
Lower Left  (-1081155.295,-2570102.995) ( 74d 0'56.06"W, 40d 8'11.55"N)
Upper Right ( 1608839.325, 1554888.755) ( 73d12'23.08"W, 41d 4'48.76"N)
Lower Right ( 1608839.325,-2570102.995) ( 73d12'48.26"W, 40d 8' 9.52"N)
Center      (  263842.015, -507607.120) ( 73d36'50.11"W, 40d36'32.68"N)
[…]
```

00610IL22L.PDF



Output Format: Projection and Parameters

- **WKT** (well-known text).
  - This is the format stored in the PDF (in particular, WKT1-ESRI).
  - **Flexible**: allows arbitrary projections and their parameters.
  - All (or nearly all) **GIS software understands this format**.
  - Generic parsing requires a schema.
  - There are some ambiguities in the way data is conveyed, particularly in the widely used WKT1.
- **PROJ**.
  - This format is used by the PROJ library.
  - **Flexible**: allows arbitrary projections and their parameters.
  - Some GIS software understands this format.
  - Parsing is straightforward, and encoding is largely unambiguous.
- ~~**GeoJSON**~~ *(infeasible)*.
  - Raised as a possibility during ACM CG 26-01.
  - Originally (2008), GeoJSON had a way to express projections in `crs` objects, but **this feature was removed from GeoJSON** (RFC 7946, 2016-08).
  - GeoJSON `crs` only provided a location to convey the projection, but for the intended purpose here (a projection with parameters), **the definition would need to appear in WKT or PROJ** anyway.
- **Hard-coded Lambert Conformal Conic**, conveying only its parameters.
  - Inflexible: **only supports this single projection method**.
  - Parameters are broken out, so they're **easy to access**.
  - **No software will understand this out of the box.** The parameters will always need to be marshalled into something else for use.

Output Format: Projection and Parameters, Examples

Each of these are equivalent.

- **WKT** (well-known text) *(WKT1-ESRI, as used in d-TPP PDFs, shown here)*.
  - `PROJCS["",GEOGCS["GCS_North_American_1983",DATUM["D_North_American_1983",SPHEROID["GRS_1980",6378137.0,298.2572221]],PRIMEM["Greenwich",0.0],UNIT["Degree",0.0174532925199433]],PROJECTION["Lambert_Conformal_Conic"],PARAMETER["False_Easting",0.0],PARAMETER["False_Northing",0.0],PARAMETER["Central_Meridian",-73.6931666666667],PARAMETER["Standard_Parallel_1",44.6666666666667],PARAMETER["Standard_Parallel_2",39.3333333333333],PARAMETER["Latitude_Of_Origin",40.7253055555556],UNIT["Inch",0.0254000508001]]`
- **PROJ** *(PROJ4 shown here)*.
  - `+proj=lcc +lat_0=40.7253055555556 +lon_0=-73.6931666666667 +lat_1=44.6666666666667 +lat_2=39.3333333333333 +x_0=0 +y_0=0 +datum=NAD83 +units=us-in +no_defs=True`
- **Hard-coded Lambert Conformal Conic**, conveying only its parameters.
  - Projection: Lambert Conformal Conic
  - Datum: NAD83
  - Standard Parallels: 44.6666666666667, 39.3333333333333
  - Origin (Latitude, Longitude): 40.7253055555556, -73.6931666666667

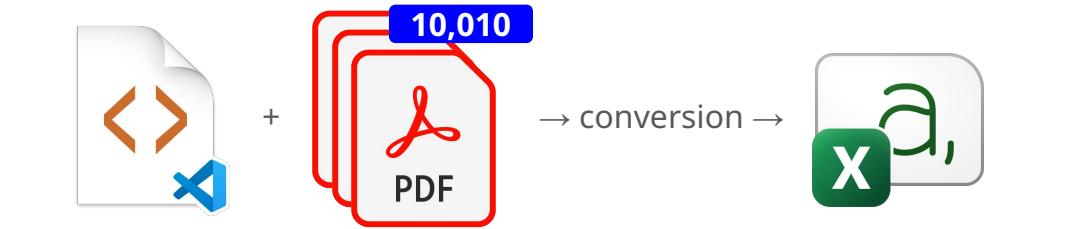
Output Format: CSV

Benefits

- This is exactly what the recommendation document requests.
- It's **easy to parse**.

Drawbacks

- **There's no standard** for georeferencing data in CSVs, so whatever is produced would be "invented".
- Although easy to parse, all users would require **custom tooling** to harness and use the data.
- ***Inflexible:*** CSV does not lend itself to the rich set of possible transforms, so the designed solution would necessarily be tied to the details of today's georeferenced charts.
- In the example that follows, unchanging parameters (projection method, datum/ellipsoid, false easting/northing) are omitted. These **must be documented**.
- *Upstream, FAA:* It's another file to produce and host somewhere for distribution.
- It's another file for users to have to obtain and match up with the data.



Output Format: CSV, converter pseudocode

- **Write the CSV header**, giving column names.
- For each `<record>` in d-TPP_Metafile.xml with `<chart_code>` = `"IAP"` and `<useraction>` ≠ `"D"` (deleted):
  - **Open the chart PDF**, using the filename from `<pdf_name>`.
  - If the PDF is a Geospatial PDF:
    - Assert that the coordinate reference system uses the Lambert Conformal Conic projection, datum NAD83, and no false easting/northing.
    - Note **projection parameters**: two standard parallels and origin (latitude, longitude).
    - Determine the **projected Cartesian (x, y) coordinates of the page corners** as distances from the projection's origin.
    - Apply the projection's reverse transformation to convert those projected Cartesian (x, y) corner coordinates to **geographic (latitude–longitude) coordinates**.
    - **Write an output record** for the chart, containing its projection parameters and corner geographic coordinates.

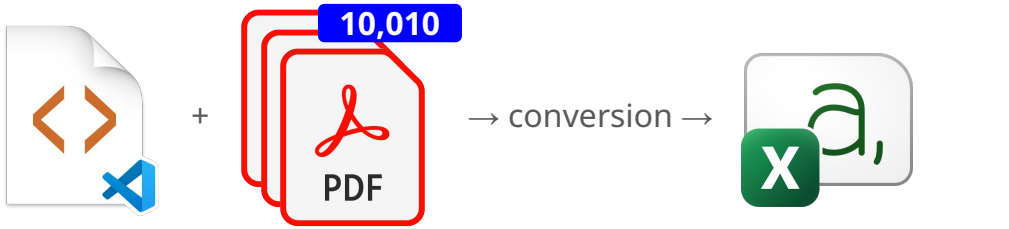
Don’t Panic!

- Some code examples follow.
  - Don’t touch that dial! **You don’t have to be a software engineer** to follow along.
  - The samples are functionally equivalent to the pseudocode, but directly usable in practice.
  - I’m showing code to make clear **how easy it is to access the data…**
  - **…and how little code it takes**. Each example fits on a single slide.
- The examples use rasterio.
  - An open-source Python library wrapping GDAL.
  - Installable from PyPI rasterio package (`pip install rasterio`).
  - Documentation: rasterio.readthedocs.io.
  - BSD-licensed source code available at https://github.com/rasterio/rasterio.
- The source code for these examples is available at https://github.com/markmentovai/faa_tpp_iap_georef/ under an Apache 2.0 license.

Output Format: CSV, converter Python code

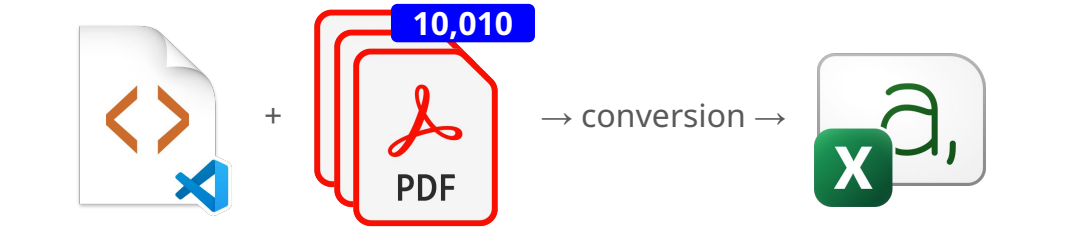
```python
def faa_tpp_iap_georef_csv(tpp_dir: os.PathLike[str] | str, csv_file: typing.TextIO) -> None:
    csv_writer = csv.writer(csv_file)

    # Write the CSV header.
    csv_writer.writerow(('filename', 'sp_lat_1', 'sp_lat_2', 'origin_lat', 'origin_lon', 'tl_lat', 'tl_lon', 'bl_lat', 'bl_lon',
                         'tr_lat', 'tr_lon', 'br_lat', 'br_lon'))

    # Scan d-TPP_Metafile.xml for charts. Only look at IAPs, and don't look at anything that's been deleted.
    metafile = xml.etree.ElementTree.parse(os.path.join(tpp_dir, 'd-TPP_Metafile.xml'))
    for chart_el in metafile.iterfind('./state_code/city_name/airport_name/record/chart_code[.="IAP"]/../useraction[.!="D"]/..'):
        pdf_name_el, = chart_el.iterfind('./pdf_name')
        pdf_name = pdf_name_el.text
        assert isinstance(pdf_name, str)
        with warnings.catch_warnings(), rasterio.Env(GDAL_PDF_DPI=300):
            warnings.simplefilter('error', rasterio.errors.NotGeoreferencedWarning)  # Allow non-georeferenced plates to be skipped.
            try:
                with rasterio.open(os.path.join(tpp_dir, pdf_name)) as rio:
                    crs_dict = rio.crs.to_dict()  # In production, assert to ensure Lambert Conformal Conic and NAD83/WGS84.

                    # Corner coordinates. "xy" are Cartesian, distances from the origin.
                    xy_tl = rio.xy(0, 0, offset='ul')
                    xy_bl = rio.xy(rio.height - 1, 0, offset='ll')
                    xy_tr = rio.xy(0, rio.width - 1, offset='ur')
                    xy_br = rio.xy(rio.height - 1, rio.width - 1, offset='lr')

                    # Corner coordinates. "ll" are geographic (latitude/longitude).
                    ll_wgs84 = rasterio.warp.transform(rio.crs, crs_dict['datum'],  # 'NAD83', or use 'WGS84' or 'EPSG:4326'
                                                       (xy_tl[0], xy_bl[0], xy_tr[0], xy_br[0]), (xy_tl[1], xy_bl[1], xy_tr[1], xy_br[1]))
                    ll_tl, ll_bl, ll_tr, ll_br = ((ll_wgs84[1][i], ll_wgs84[0][i]) for i in range(4))

                    # Write this chart's information to the CSV output. 7 decimal places is .00036" resolution, better than dd°mm'ss.sss", ≤1.1cm.
                    csv_writer.writerow((pdf_name, *(float(crs_dict[k]) for k in ('lat_1', 'lat_2', 'lat_0', 'lon_0',)),
                                         *('%.7f' % f for f in (*ll_tl, *ll_bl, *ll_tr, *ll_br))))
            except rasterio.errors.NotGeoreferencedWarning:
                pass
```

Output Format: CSV, performing conversion




```
% unzip -d tpp_2608 -q DDTPPA_260806.zip
% unzip -d tpp_2608 -q DDTPPB_260806.zip
% unzip -d tpp_2608 -q DDTPPC_260806.zip
% unzip -d tpp_2608 -q DDTPPD_260806.zip
% unzip -d tpp_2608 -q DDTPPE_260806.zip

% python3 faa_tpp_iap_georef_csv.py tpp_2608 IAP_Georeferencing.csv
[1 minute 38 seconds later…]
```

Output Format: CSV, output



Raw CSV

```
filename,sp_lat_1,sp_lat_2,origin_lat,origin_lon,tl_lat,tl_lon,bl_lat,bl_lon,tr_lat,tr_lon,br_lat,br_lon
01244IYLY23.PDF,54.6666666666667,49.3333333333333,51.8835833333333,-176.642472222222,52.3603351,-177.0224124,51.4176736,-177.0145312,52.3593028,-176.0184025,51.4166626,-176.0313472
01244IZLZ23.PDF,54.6666666666667,49.3333333333333,51.8835833333333,-176.642472222222,52.3603351,-177.0224124,51.4176736,-177.0145312,52.3593028,-176.0184025,51.4166626,-176.0313472
[…]
00610IL13L.PDF,44.6666667,39.3333333,40.6946389,-73.8686389,41.0696208,-74.2562796,40.1253823,-74.2507098,41.0694826,-73.4425561,40.1252460,-73.4486782
00610IL22L.PDF,44.6666666666667,39.3333333333333,40.7253055555556,-73.6931666666667,41.0807867,-74.0202725,40.1365428,-74.0155717,41.0802124,-73.2064118,40.1359769,-73.2134068
00610IL22R.PDF,44.6666666666667,39.3333333333333,40.7263888888889,-73.7098888888889,41.0915435,-74.0377594,40.1472971,-74.0330468,41.0909742,-73.2237634,40.1467361,-73.2307505
[…]
06048N7.PDF,10.0,9.0,9.49997222222222,138.097583333333,9.9245549,137.7872250,8.9772341,137.7880769,9.9245528,138.4102771,8.9772320,138.4094189
06048N25.PDF,10.0,9.0,9.49944444444444,138.088472222222,9.9163341,137.7781213,8.9690130,137.7789731,9.9163320,138.4011585,8.9690109,138.4003003
```

Output Format: CSV, output

Tabular view, as imported into Google Sheets (or any spreadsheet program)

filename	sp_lat_1	sp_lat_2	origin_lat	origin_lon	tl_lat	tl_lon	bl_lat	bl_lon	tr_lat	tr_lon	br_lat	br_lon
01244IYLY23.PDF	54.666667	49.333333	51.883583	-176.642472	52.360335	-177.022412	51.417674	-177.014531	52.359303	-176.018403	51.416663	-176.031347
01244IZLZ23.PDF	54.666667	49.333333	51.883583	-176.642472	52.360335	-177.022412	51.417674	-177.014531	52.359303	-176.018403	51.416663	-176.031347
00610IL4L.PDF	44.666667	39.333333	40.562250	-73.832194	41.014891	-74.201373	40.070665	-74.196073	41.014622	-73.388338	40.070400	-73.394710
00610IL4R.PDF	44.666667	39.333333	40.586778	-73.799528	40.992687	-74.211031	40.048469	-74.205125	40.992724	-73.398272	40.048505	-73.404031
00610IL13L.PDF	44.666667	39.333333	40.694639	-73.868639	41.069621	-74.256280	40.125382	-74.250710	41.069483	-73.442556	40.125246	-73.448678
00610IL22L.PDF	44.666667	39.333333	40.725306	-73.693167	41.080787	-74.020273	40.136543	-74.015572	41.080212	-73.206412	40.135977	-73.213407
00610IL22R.PDF	44.666667	39.333333	40.726389	-73.709889	41.091544	-74.037759	40.147297	-74.033047	41.090974	-73.223763	40.146736	-73.230751
00610IL31L.PDF	45.000000	33.000000	40.590917	-73.684111	40.925960	-74.090196	39.977519	-74.084685	40.925950	-73.274971	39.977509	-73.280523
00610IL31R.PDF	44.666667	39.333333	40.597417	-73.657389	40.993392	-74.062250	40.049174	-74.056439	40.993381	-73.249482	40.049163	-73.255336
06048N7.PDF	10.000000	9.000000	9.499972	138.097583	9.924555	137.787225	8.977234	137.788077	9.924553	138.410277	8.977232	138.409419
06048N25.PDF	10.000000	9.000000	9.499444	138.088472	9.916334	137.778121	8.969013	137.778973	9.916332	138.401159	8.969011	138.400300

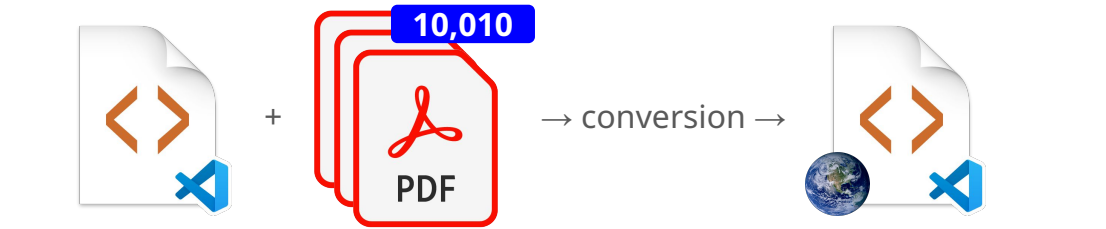
Output Format: d-TPP_Metafile.xml

Benefits

- *Upstream, FAA:* **This file is already being distributed**.
- Users that process FAA IAP charts are **probably already consuming this file**, and are **probably already familiar with it**.
- The “X” in XML is “eXtensible”: it’s possible to **augment the format** without disturbing existing consumers.
- Data can be highly **structured**, **tailored** to the application.
- It’s possible to **encode the projection and its parameters fully** (for example, as WKT).
- It’s also possible to provide a **decoded form** specific to the Lambert Conformal Conic projection in use.

Drawbacks

- There’s no standard for georeferencing data in XML, so whatever is produced would be “invented”. *(Same as CSV.)*
- Although easy to parse, all users would require custom tooling to harness and use the data. *(Same as CSV.)*
- It’s not as easy as CSV to look at in tabular form.



d-TPP_Metafile.xml (existing format)

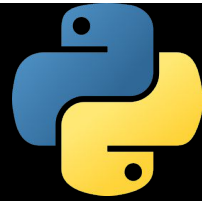
```
<?xml version="1.0" encoding="UTF-8"?>
<digital_tpp cycle="2608" from_edate="0901Z  08/06/26" to_edate="0901Z  09/03/26">
  <state_code ID="NY" state_fullname="NEW YORK">
    <city_name ID="NEW YORK" volume="NE-2">
      <airport_name ID="JOHN F KENNEDY INTL" military="N" apt_ident="JFK" icao_ident="KJFK" alnum="610">
        <record>
          <chartseq>50750</chartseq>
          <chart_code>IAP</chart_code>
          <chart_name>ILS OR LOC RWY 22L</chart_name>
          <useraction></useraction>
          <pdf_name>00610IL22L.PDF</pdf_name>
          <cn_flg>N</cn_flg>
          <cnsection></cnsection>
          <cnpage></cnpage>
          <bvsection></bvsection>
          <bvpage>233</bvpage>
          <procuid>2831</procuid>
          <two_colored>N</two_colored>
          <civil>C</civil>
          <faanfd18></faanfd18>
          <copter></copter>
          <amdtnum>26</amdtnum>
          <amdtdate>12/31/2020</amdtdate>
        </record>
```

Output Format: d-TPP_Metafile.xml, converter pseudocode

~~Subtractions~~ and additions compared to the CSV converter.

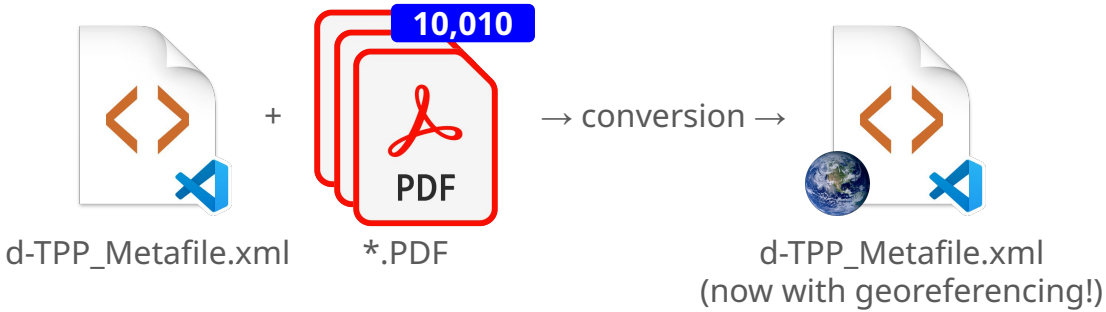
- ~~**Write the CSV header,** giving column names.~~
- For each `<record>` in d-TPP_Metafile.xml with `<chart_code>` = `"IAP"` and `<useraction>` ≠ `"D"`:
  - **Open the chart PDF**, using the filename from `<pdf_name>`.
  - If the PDF is a Geospatial PDF:
    - Optional: Assert that the coordinate reference system uses the Lambert Conformal Conic projection, datum NAD83, and no false easting/northing.
    - Note **projection parameters**: two standard parallels and origin (latitude, longitude).
    - Determine the **projected Cartesian (x, y) coordinates of the page corners** as distances from the projection's origin.
    - Apply the projection's reverse transformation to convert those projected Cartesian (x, y) corner coordinates to **geographic (latitude–longitude) coordinates**.
    - ~~**Write an output record** for the chart, containing its projection parameters and corner geographic coordinates.~~
    - **Add a `<georeferencing>` element to the `<record>`**, containing the chart's geographic coordinate system definition (WKT), projection parameters, and corner geographic coordinates as control points.
- Output the XML file, including added `<georeferencing>` elements.

Output Format: d-TPP_Metafile.xml, converter Python code



```python
def _xml_el(
    tag: str, attrib: dict[str, str] = {}, text: str | None = None, children: typing.Iterable[xml.etree.ElementTree.Element] = ()
) -> xml.etree.ElementTree.Element:
    el = xml.etree.ElementTree.Element(tag, attrib=attrib)
    el.text = text
    el.extend(children)
    return el

def faa_tpp_iap_georef_xml(tpp_dir: os.PathLike[str] | str, xml_write_file: typing.TextIO) -> None:
    # […]

                    # (From the CSV example,) for each IAP chart that hasn't been deleted and that is georeferenced:
                    ll_tl, ll_bl, ll_tr, ll_br = ((ll_wgs84[1][i], ll_wgs84[0][i]) for i in range(4))

                    # Insert this chart's information into the XML structure.
                    chart_el.append(
                        _xml_el('georeferencing', children=(
                            _xml_el('coordinate_reference_system',
                                    attrib={'format': 'wkt', 'version': 'WKT1_ESRI'}, text=rio.crs.to_wkt(version='WKT1_ESRI')),
                            _xml_el('projection', attrib={'type': crs_dict['proj'], 'datum': crs_dict['datum']}, children=(
                                _xml_el('standard_parallel', attrib={'latitude': str(float(crs_dict['lat_1']))}),
                                _xml_el('standard_parallel', attrib={'latitude': str(float(crs_dict['lat_2']))}),
                                _xml_el('origin', attrib={'latitude': str(float(crs_dict['lat_0'])), 'longitude': str(float(crs_dict['lon_0']))}),
                            )),
                            _xml_el('control_points', children=(
                                _xml_el('control_point',
                                        attrib={'latitude': '%.7f' % lat, 'longitude': '%.7f' % lon, 'x': str(float(x)), 'y': str(float(y))})
                                for (lat, lon), (x, y) in zip((ll_tl, ll_bl, ll_tr, ll_br), ((0, 1), (0, 0), (1, 1), (1, 0)))),
                            ),
                        )))

            except rasterio.errors.NotGeoreferencedWarning:
                pass

    # Write the XML output.
    xml.etree.ElementTree.indent(metafile)
    metafile.write(xml_write_file, encoding='unicode', xml_declaration=True, short_empty_elements=False)
```

Output Format: d-TPP_Metafile.xml, performing conversion




```
% python3 faa_tpp_iap_georef_xml.py tpp_2608 d-TPP_Metafile_Georeferenced.xml
[1 minute 52 seconds later…]
```

Output Format: d-TPP_Metafile.xml, output, now with georeferencing!

```
<?xml version='1.0' encoding='UTF-8'?>
<digital_tpp cycle="2608" from_edate="0901Z  08/06/26" to_edate="0901Z  09/03/26">
  <state_code ID="NY" state_fullname="NEW YORK">
    <city_name ID="NEW YORK" volume="NE-2">
      <airport_name ID="JOHN F KENNEDY INTL" military="N" apt_ident="JFK" icao_ident="KJFK" alnum="610">
        <record>
[…]
          <chart_name>ILS OR LOC RWY 22L</chart_name>
[…]
          <pdf_name>00610IL22L.PDF</pdf_name>
[…]
          <amdtdate>12/31/2020</amdtdate>
          <georeferencing>
            <coordinate_reference_system format="wkt" version="WKT1_ESRI">PROJCS["",GEOGCS["GCS_North_American_1983",DATUM["D_North_
American_1983",SPHEROID["GRS_1980",6378137.0,298.2572221]],PRIMEM["Greenwich",0.0],UNIT["Degree",0.0174532925199433]],PROJECTION["La
mbert_Conformal_Conic"],PARAMETER["False_Easting",0.0],PARAMETER["False_Northing",0.0],PARAMETER["Central_Meridian",-73.693166666666
7],PARAMETER["Standard_Parallel_1",44.6666666666667],PARAMETER["Standard_Parallel_2",39.3333333333333],PARAMETER["Latitude_Of_Origin
",40.7253055555556],UNIT["Inch",0.0254000508001]]</coordinate_reference_system>
            <projection type="lcc" datum="NAD83">
              <standard_parallel latitude="44.6666666666667"></standard_parallel>
              <standard_parallel latitude="39.3333333333333"></standard_parallel>
              <origin latitude="40.7253055555556" longitude="-73.6931666666667"></origin>
            </projection>
            <control_points>
              <control_point latitude="41.0807867" longitude="-74.0202725" x="0.0" y="1.0"></control_point>
              <control_point latitude="40.1365428" longitude="-74.0155717" x="0.0" y="0.0"></control_point>
              <control_point latitude="41.0802124" longitude="-73.2064118" x="1.0" y="1.0"></control_point>
              <control_point latitude="40.1359769" longitude="-73.2134068" x="1.0" y="0.0"></control_point>
            </control_points>
          </georeferencing>
        </record>
```

Technique: Use the Data Directly

Now that you know how to access the data in the PDFs, it's also possible to use it directly from there without any intermediate steps or files.

Benefits

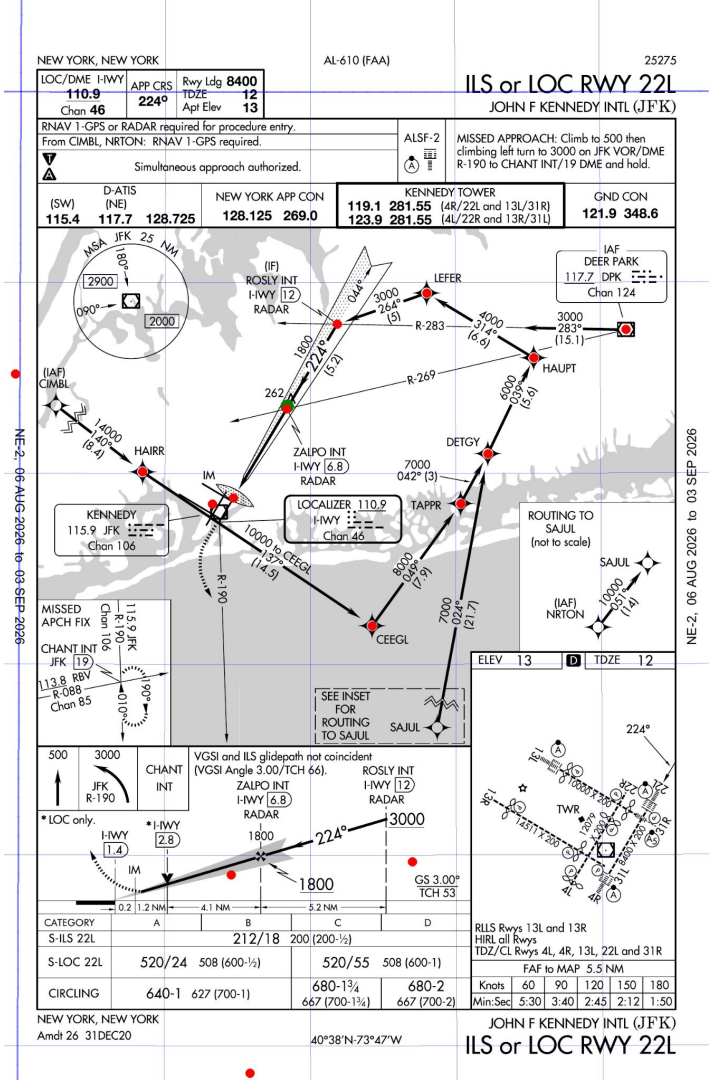
- No additional files to produce or distribute.

Drawbacks

- Each new application needs to become aware of Geospatial PDFs.

00610IL22L.PDF FAACIFP18 00610IL22L.png (marked up)

```
% unzip -d cifp_2608 CIFP_260806.zip

% python3 faa_tpp_iap_georef_chart.py \
      --a424-file=cifp_2608/FAACIFP18 \
      --a424-proc=KJFKK6FI22L \
      --grid \
      00610IL22L.PDF
```



DIY (Do It Yourself): What if you can't or don't want to use GDAL?

- *Are you sure?*
  - **Your GIS software may already use GDAL** under the hood, or make GDAL available.
- *Yes, you're sure.*
  - *Downstream, data suppliers:* Qualifying a large and complex library under FAA AC 20-153□ and RTCA DO-200□ may be…
    - difficult,
    - time-consuming, or
    - expensive.
  - I'm a fan of open-source software, but you might not be.
    - Uncertain support might be a problem for business-critical processes, particularly if you're not willing or able to build and maintain in-house expertise.
- The GeoTIFF, CSV, d-TPP_Metafile.xml conversion examples and the direct-use example so far have all relied on GDAL.
- Can you **write your own software to extract georeferencing information** directly from Geospatial PDFs?
  - Yes!
  - What do you need in order to do this?
    - A way to **access the structure of a PDF file**. You may already have a library for this, but if not, it's possible to DIY it too.
    - A willingness to do a little math to handle the projection and other transformations. It's not much, and examples are readily available.
  - This isn't theoretical, I've done it.
    - *(And it runs in $^{1}/_{30}$ the time that GDAL does!)*

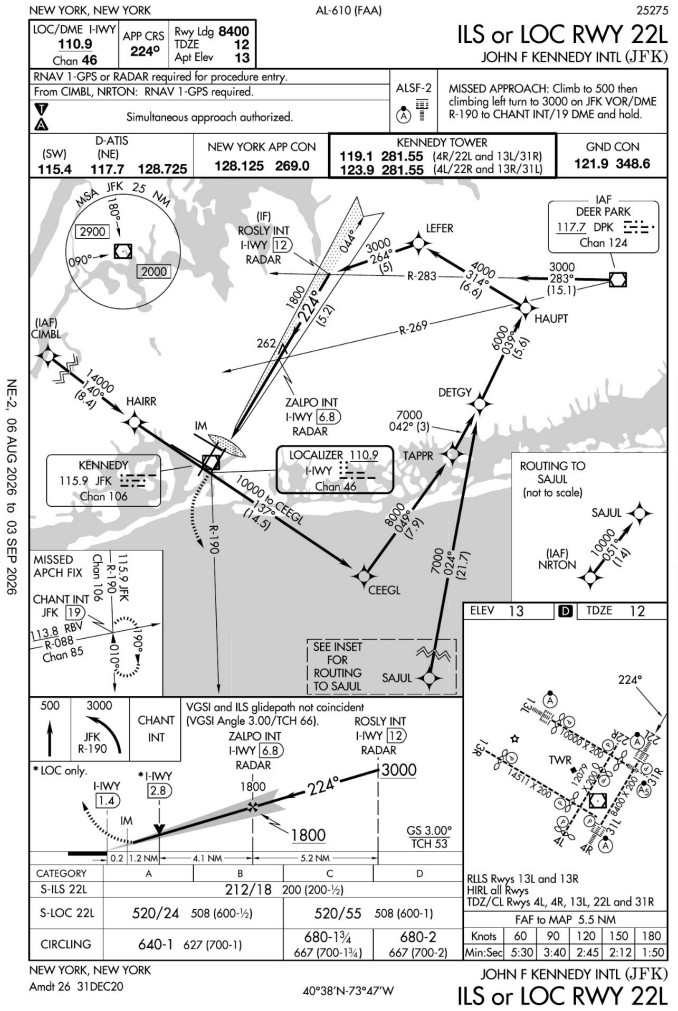
DIY: Accessing Georeferencing Data

PDF

00610IL22L.PDF

```
%PDF-1.4
8 0 obj
<</Metadata 4 0 R/Names 5 0 R/Outlines 2 0 R/Pages 1 0 R/Type/Catalog>>
endobj
10 0 obj
<</CropBox[0 0 387.36 594]/MediaBox[0 0 387.36 594]/VP[<</BBox[9.18 2.628
 378.18 591.372]/Measure<</Bounds[0 0 1 0 1 1 0 1 0 0]/GCS<</EPSG 0/Type/
PROJCS/WKT(PROJCS["",GEOGCS["GCS_North_American_1983",DATUM["D_North_Amer
ican_1983",SPHEROID["GRS_1980",6378137.000,298.25722210]],PRIMEM["Greenwi
ch",0],UNIT["Degree",0.017453292519943295]],PROJECTION["Lambert_Conformal
_Conic"],PARAMETER["False_Easting",0.000],PARAMETER["False_Northing",0.00
0],PARAMETER["Central_Meridian",-73.693166666666671],PARAMETER["Latitude_O
f_Origin",40.72530555555561],PARAMETER["Standard_Parallel_1",44.666666666
66671],PARAMETER["Standard_Parallel_2",39.333333333333331],UNIT["Inch",0.0
254000508000010]])>>/GPTS[40.23453277014 -73.97048533612 40.23410083011 -73
.30825723326 40.98281267452 -73.30381398641 40.98324968235 -73.9231094390
7]/LPTS[0.1 0.1 0.9 0.1 0.9 0.9 0.1 0.9]/PDU[(MI)(SQMI)(DEG)]/Subtype/GEO
/Type/Measure>>>>]>>
endobj
1 0 obj
<</Count 1/Kids[10 0 R]/Type/Pages>>
endobj
trailer <</Root 8 0 R>>
%%EOF
```

start here

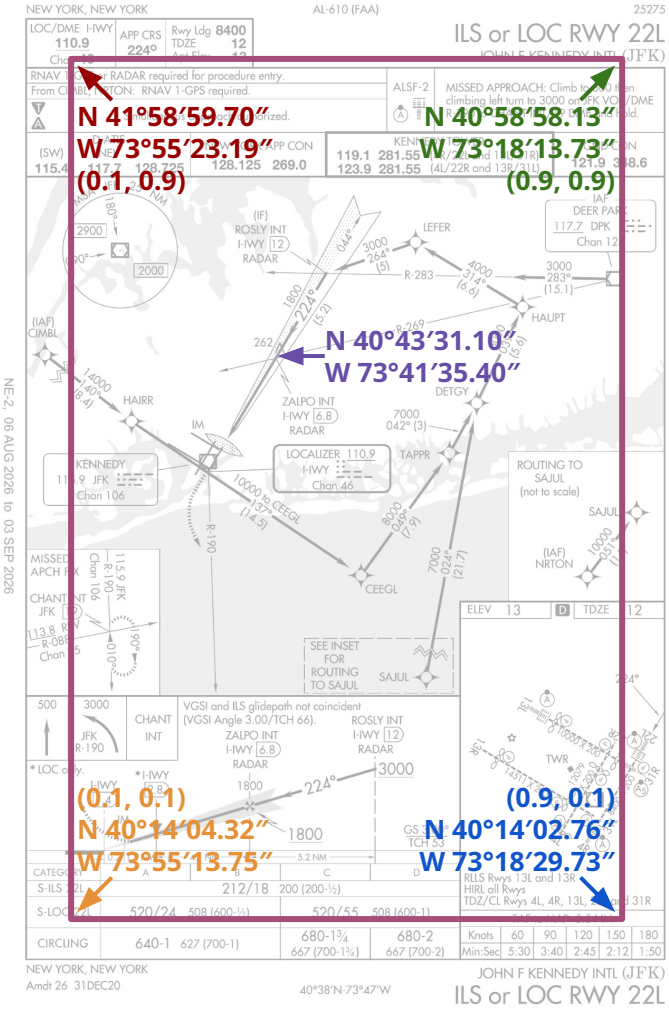


DIY: Control Point Extraction

PDF

00610IL22L.PDF


```
%PDF-1.4
8 0 obj
<</Metadata 4 0 R/Names 5 0 R/Outlines 2 0 R/Pages 1 0 R/Type/Catalog>>
endobj
10 0 obj
<</CropBox[0 0 387.36 594]/MediaBox[0 0 387.36 594]/VP[<</BBox[9.18 2.628
 378.18 591.372]/Measure<</Bounds[0 0 1 0 1 1 0 1 0 0]/GCS<</EPSG 0/Type/
PROJCS/WKT(PROJCS["",GEOGCS["GCS_North_American_1983",DATUM["D_North_Amer
ican_1983",SPHEROID["GRS_1980",6378137.000,298.25722210]],PRIMEM["Greenwi
ch",0],UNIT["Degree",0.017453292519943295]],PROJECTION["Lambert_Conformal
_Conic"],PARAMETER["False_Easting",0.000],PARAMETER["False_Northing",0.00
0],PARAMETER["Central_Meridian",-73.69316666666671],PARAMETER["Latitude_O
f_Origin",40.72530555555561],PARAMETER["Standard_Parallel_1",44.666666666
66671],PARAMETER["Standard_Parallel_2",39.33333333333331],UNIT["Inch",0.0
2540005080010]])>>/GPTS[40.23453277014 -73.92048533612 40.23410083011 -73
.30825723326 40.98281267452 -73.30381398641 40.98324968235 -73.9231094390
7]/LPTS[0.1 0.1 0.9 0.1 0.9 0.9 0.1 0.9]/PDU[(MI)(SQMI)(DEG)]/Subtype/GEO
/Type/Measure>>>>]>>
endobj
1 0 obj
<</Count 1/Kids[10 0 R]/Type/Pages>>
endobj
trailer <</Root 8 0 R>>
%%EOF
```

DIY: Control Point Conversion

00610IL22L.PDF

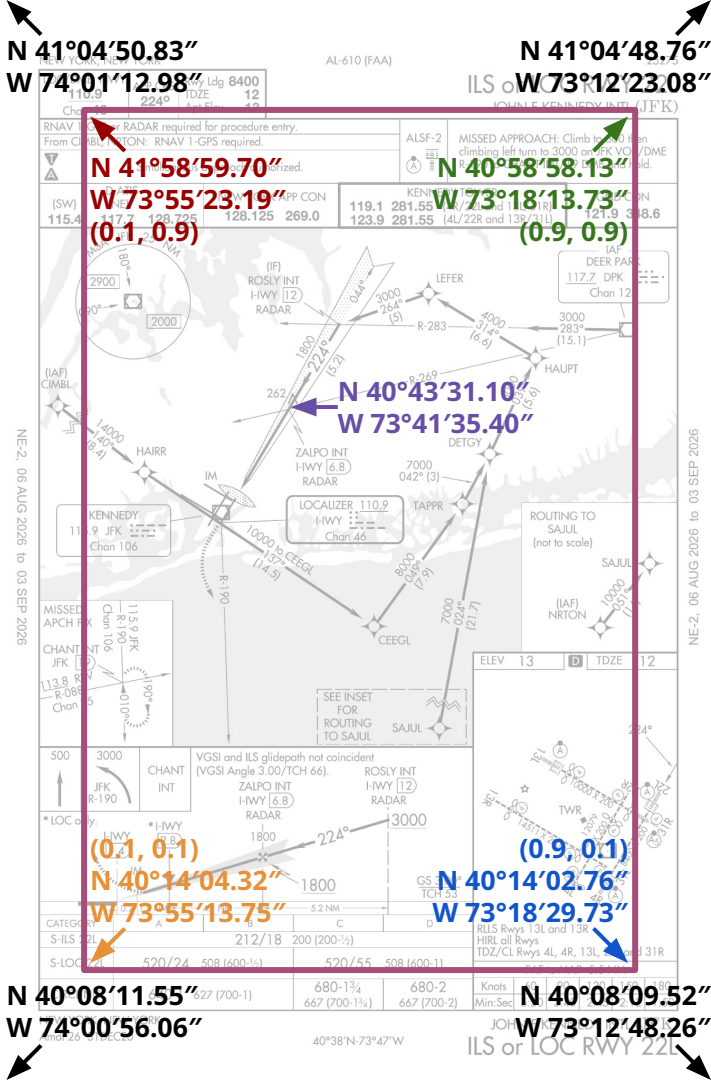
The magenta rectangle is arbitrary and not very useful! The page's **corner coordinates** would be better, like the previous (GDAL-using) samples provide, and like the recommendation document suggests.

- Set up a Lambert Conformal Conic projection transformation based on the parameters in the PDF (PROJCS).
- Apply the projection's forward transformation to **convert the geographic (latitude–longitude) coordinates** of the magenta box' corners **into projected Cartesian (x, y) coordinates** relative to the projection origin.
- **Determine the projected Cartesian (x, y)** page **corner coordinates** as distances from the projection origin.
- Apply the projection's reverse transformation to **convert those projected Cartesian (x, y) corner coordinates to geographic (latitude–longitude) coordinates**.

It's easier than it sounds!

References for Lambert Conformal Conic projection transformations:

- EPSG Guidance Note 7-2, §3.4.1.1, Lambert Conic Conformal (2SP).
- EPSG Method 9802, Lambert Conic Conformal (2SP).
- USGS Map Projections: A Working Manual (1987), §15, Lambert Conformal Conic.

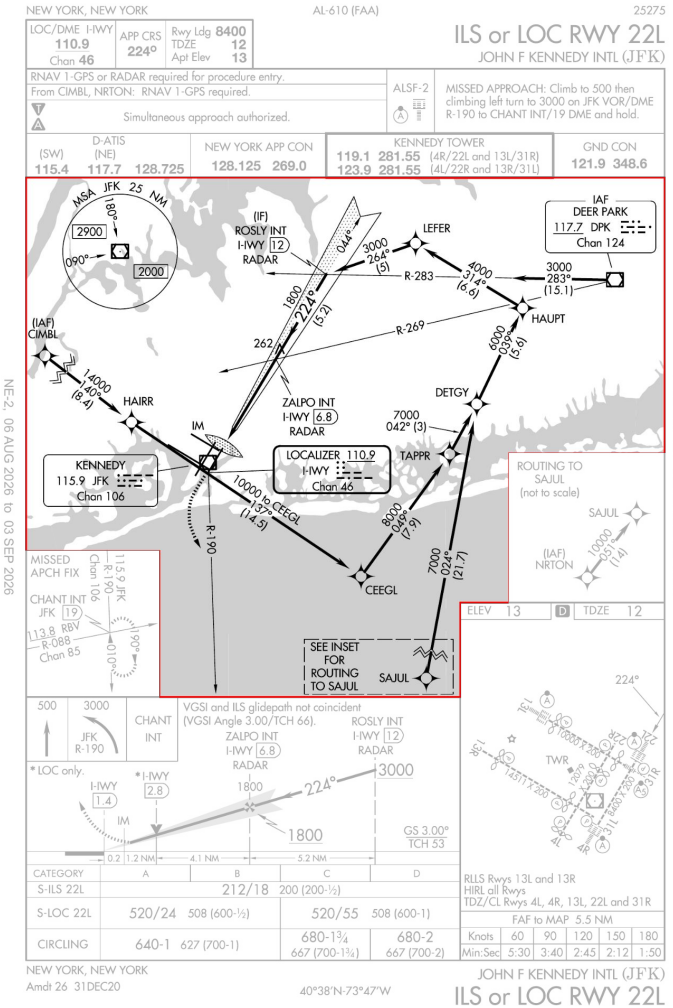


Neatline Extraction

The recommendation document also seeks “ideally, coordinates of the edges of the planview of the chart.”

00610IL22L.PDF

- Geospatial PDF *does* provide a way to store this information (**Bounds**), but it’s set to essentially the entire page in FAA instrument approach chart PDFs: `/Bounds[0 0 1 0 1 1 0 1 0 0]`.
- Technique: **recover the dimensions of the planview** from these PDFs in software, by interpreting the PDFs’ drawing commands.
- This can account for and excludes most **insets** (CHANT and NRTON), but not **scale breaks** (CIMBL and SAJUL).
- I’ve focused this presentation on extracting structured data, so I will not demonstrate this technique in the presentation.
- The image on this slide is intended to illustrate the concept, and was not produced by software.
- *Output format guidance.* Illustrating possibilities for future extensibility:
  - It’s possible to **carry this information directly in a Geospatial PDF**.
  - It can also be easily carried in a **further extension to the XML** structure.
  - **CSV is a poor fit for this information**, in particular because the number of points in the shape is variable.



Data Quirks

Oddities in TPP 2608:

- In addition to 10,010 IAPs, there are **4 departure chart pages and 1 arrival chart page** that are georeferenced. **The georeferencing is incorrect**.
  - KPHX BROAK1 (RNAV) departure: 00322BROAK.PDF (procedure 1806, chart 18144).
  - KPHX ECLPS1 (RNAV) departure: 00322ECLPS.PDF (procedure 1806, chart 18144).
  - KPHX FYRBD1 (RNAV) departure: 00322FYRBD.PDF (procedure 1806, chart 18144).
  - KPHX MRBIL1 (RNAV) departure, page 2: 00322MRBIL_C.PDF (procedure 1806, chart 18144, no plan view!).
  - KONT SCBBY2 (RNAV) arrival, page 2: 00965SCBBY_C.PDF (procedure 1802, chart 18032).
- All georeferenced charts are “true north up” (not rotated) except for one. GDAL handles this well, but it would take more effort for handmade GIS tools to accommodate this.
  - NSTU/PPG approach ILS or LOC 5: 05018IL5.PDF (procedure 2108, chart 26134), the plan view is **rotated 0.16° clockwise**.

Source Code Availability

The sample code included in this presentation, along with its output, is available under the Apache 2.0 license at https://github.com/markmentovai/faa_tpp_iap_georef/.